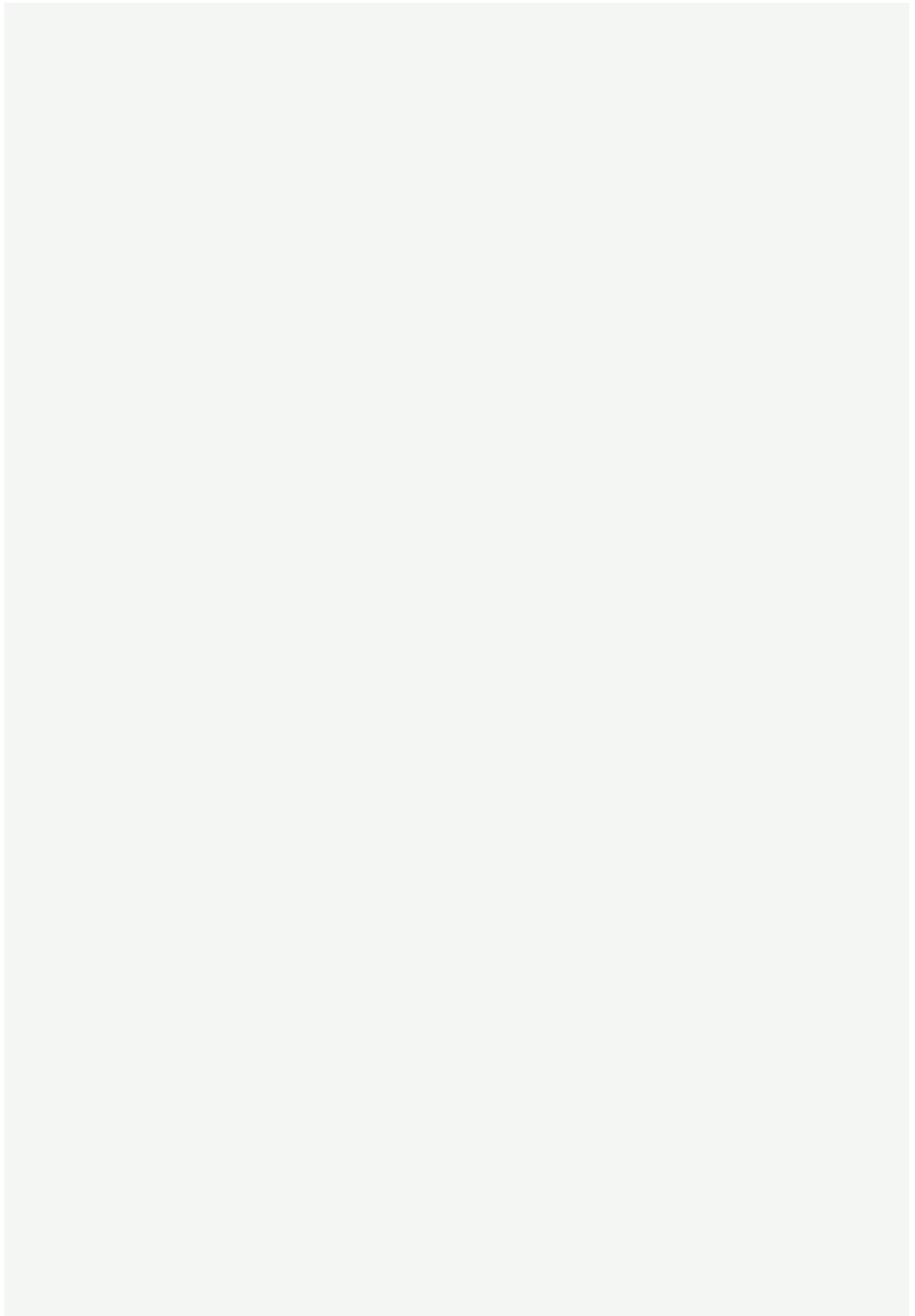




2025



Introduction

Kumaraguru Institutions is a multi-campus educational institution in Tamil Nadu, South India, offering undergraduate, postgraduate, and research programs in Engineering, Technology, Management, Agriculture, Liberal Arts, and Science. Guided by its motto, "Character is Life," the institution is committed to academic excellence, holistic development, and nurturing responsible individuals with integrity and purpose. Through experiential learning, innovative research, and strong community engagement, Kumaraguru fosters a culture of inquiry and social responsibility, empowering students to address real-world challenges and contribute meaningfully to society.



Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological

Introduction

Kumaraguru Institutions recognize the imperative to address climate change and integrate environmental sustainability into campus operations. This report encapsulates our Greenhouse Gas (GHG) inventory, providing a transparent account of our carbon footprint and resource utilization.

The scope of this report covers all major operational aspects, including energy consumption, transportation, and direct emissions from stationary sources. By quantifying these impacts, we lay the groundwork for targeted reduction strategies and continuous improvement.

Strategic Objectives

- Establish a robust baseline for GHG emissions and energy consumption.
- Identify key emission hotspots to prioritize mitigation interventions.
- Monitor the effectiveness of ongoing sustainability initiatives, such as renewable energy integration.
- Foster a culture of environmental stewardship among students, faculty, and staff.

Highlights

KUMARAGURU
Institutions

EMISSION PROFILE

Total GHG
Emissions

1753.80 tCO2e

Emissions

Emissions (DG +
Transport)

389.1 tCO2e

ILE

Emissions (Electricity) **1364.7** tCO2e

Em
Tra

Emissions (LPG - Canteen) **0.00** tCO2e

ENERGY INDICATORS

Total Energy
Consumption

5250515 Kwh

Energy

tCO2e

Em
Ca

Grid Energy
Consumption

1877147 Kwh

Em
Ca

Renewable Energy
Consumption

3373368 Kwh

TO

Renewable Energy Share

64.2 %

Highlights

KUMARAGURU
Institutions

TRANSPORT INDICATORS (FLEET)

Total Fuel
Consumption

126034 Litres

Total Petrol
Consumption

8954.11 Litres

ILE

Total Diesel
Consumption

117080 Litres

Emissions (DG +
Tra

337.6

Total Fleet Emissions

337.6 tCO₂e

Highlights

KUMARAGURU
Institutions

CLIMATE ACTION INDICATORS

Avoided
Emissions

2452.4 tCO2e

Per Capita
Emissions

0.000 tCO2e/person

FIL

En

Executive Summary

KUMARAGURU
Institutions

High-level overview of carbon performance for Carbon Emissions Report - 2025.

Highlights



Performance Overview EMISSION PROFILE

During this reporting period, the campus generated 1753.80 tCO₂e of gross carbon emissions. Through our renewable energy initiatives, we successfully avoided 2452.44 tCO₂e, resulting in a net carbon impact of -698.64 tCO₂e. We remain committed to our long-term decarbonization pathway.

Total GHG
Emissions

Emissions (DG +
transport)

389.1
tCO₂e

Emissions
(Electricity)

1364.7
tCO₂e

Emissions (LPG -

158.96

CARBON STORY

1912.76

tCO₂e

Gross GHG
Emissions

Net GHG
Emissions

Avoided
Emissions

1753.8 tCO₂e

IEE

-698.6 tCO₂e

2452.4 tCO₂e

STRATEGIC RECOMMENDATIONS

Total Energy
Consumption

- Accelerate the transition of internal campus mobility to Electric Vehicles (EVs) to reduce Scope 1 transport emissions.

Total Energy
Consumption

107,114.7
Kwh

- Expand rooftop solar capacity to push the renewable energy mix beyond the current baseline.

Renewable Energy
Consumption

3373368
Kwh

- Implement automated BMS (Building Management Systems) for high-load commercial blocks to optimize HVAC consumption.

52,30,515
Kwh

Renewable
Energy Share

63.6%

Scope 1 Story

KUMARAGURU
Institutions

Direct Emissions Overview

Scope 1 emissions generated 389.11 tCO₂e. Transport remains the dominant emission source in this category. Overall Scope 1 emissions decreased by 0% compared to the previous period.



EMISSION PROFILE

Transport Deep Dive

Transport emissions decreased by 0%. Continued fleet optimization is contributing to this reduction.

Emissions (DG +
Transport)

389.1
tCO₂e

Emissions
(Electricity)

1364.7
tCO₂e

Scope 1 Composition

Chart

Er
C:

Emissions Trend

Chart



Transport Emissions (Scope 1)

337.6 tCO₂e

MONTH	PETROL (LITRES)	EMISSIONS (TCO ₂ E)	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	746.0	1.7814	9757	26.3537
FEB	483.5	1.1546	7230	19.5282
MAR	493.0	1.1773	9063	24.4792
APR	252.0	0.6018	5771	15.5875
MAY	440.0	1.0507	6435	17.3809
JUN	414.0	0.9886	9184	24.8060
JUL	853.0	2.0370	8955	24.1875
AUG	908.0	2.1683	9184	24.8060
SEP	1123.0	2.6817	8588	23.1962
OCT	976.2	2.3312	15060	40.6771
NOV	1124.6	2.6855	13629	36.8119
DEC	1140.8	2.7242	14224	38.4190

DG Set Emissions (Scope 1)

51.5 tCO₂e

MONTH	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	1588.62	4.2909
FEB	1319.34	3.5635
MAR	274.23	0.7407
APR	1273.80	3.4405
MAY	1628.22	4.3978
JUN	951.39	2.5697
JUL	214.17	0.5785
AUG	1849.32	4.9950
SEP	2959.11	7.9926
OCT	3368.64	9.0987
NOV	2162.82	5.8418
DEC	1475.43	3.9851

Scope 2 Story



Indirect Emissions Overview

Highlights

Grid electricity consumption accounted for 1364.69 tCO₂e. This represents 77.8% of total gross emissions. Scope 2 emissions fell by 0%. Continuous efforts in optimizing HVAC and lighting can help reduce this footprint.



EMISSION PROFILE

Renewable Impact

Total GHG Emissions

Renewable energy integration contributed 3373368.00 kWh, offsetting approximately 2452.44 tCO₂e of potential emissions. Our current renewable share stands at 64.2%.

1912.76

Emissions (DG + Transport)

389.1
tCO₂e

Emissions (Electricity)

1364.7
tCO₂e

Emissions (IPG - C)

152.06

Renewable Energy Mix

Chart

Grid Consumption Profile

Chart



ENERGY INDICATORS

Electricity Emissions (Scope 2)

1364.7 tCO2e

RENEWABLE ENERGY

3373368 kWh

GRID ENERGY

1877147 kWh

TOTAL CONSUMPTION

5250515 kWh

MONTH	GRID HT		GRID COMMERCIAL		GRID TEMPORARY	
	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)
JAN	155222	112.846	1023	0.744	0	0.000
FEB	194516	141.413	862	0.627	0	0.000
MAR	261397	190.036	1119	0.814	0	0.000
APR	139071	101.105	1140	0.829	0	0.000
MAY	1321	0.960	1001	0.728	0	0.000
JUN	690	0.502	1023	0.744	0	0.000
JUL	5934	4.314	1128	0.820	428	0.311
AUG	424410	308.546	1094	0.795	416	0.302
SEP	450216	327.307	1027	0.747	389	0.283
OCT	76958	55.948	1000	0.727	346	0.252
NOV	105958	77.031	963	0.700	331	0.241
DEC	46972	34.149	899	0.654	293	0.213

Emissions Comparison

KUMARAGURU
Institutions

Performance analysis vs previous reporting period

Highlights

CONSOLIDATED GHG SUMMARY

CATEGORY	CURRENT PERIOD	PREVIOUS PERIOD	% CHANGE
Total Emissions	1753.80	N/A	0.00%
Scope 1 Emissions	389.11	N/A	0.00%
Scope 2 Emissions	1364.69	N/A	0.00%
Renewable Energy (kWh)	3373368	N/A	0.00%

1912.76
Progress Summary tCO2e

Emissions (LPG -
Canteen) **158.96**
tCO2e

The comparison table above illustrates our performance trajectory against the previous baseline. Reductions in total emissions are primarily driven by continuous optimization in energy usage and sustained renewable integration.

ENERGY INDICATORS

Methodology

KUMARAGURU
Institutions

Purpose of the Report

Highlights

This report presents the Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions for the reporting period, compiled in accordance with ISO 14064-1:2018. The objective of this inventory is to measure and disclose the institution's GHG emissions, and provide reliable data to support future emissions reduction strategies.



EMISSION PROFILE

GHG Quantification Approach

The inventory has been developed using an activity-based quantification approach. Emissions were calculated by multiplying activity data (e.g., fuel consumption, electricity consumption) with appropriate emission factors derived from recognized national and international sources (IPCC, CEA, DEFRA).

Total GHG Emissions

1912.76
tCO₂e

Emissions (DG +
Transport)

389.1
tCO₂e

Emissions
(Electricity)

1364.7
tCO₂e

Organizational Boundary

The organizational boundary has been defined using the operational control approach. All facilities and operations over which Kumaraguru Institutions has operational control are included. Scope 3 emissions have been assessed at a screening level but are excluded from quantitative reporting due to the unavailability of consolidated data.

Emissions (IPCC
Canteen)

100.00
tCO₂e

Global Warming Potential (GWP)

ENERGY INDICATORS

All greenhouse gases were converted into carbon dioxide equivalent (CO₂e) using Global Warming Potential (GWP) values from the IPCC Sixth Assessment framework.



Conclusion



This Greenhouse Gas (GHG) inventory enhances the understanding of emission sources within campus operations and provides a foundation for tracking environmental performance and identifying opportunities for energy efficiency and emission reduction. The institution intends to progressively improve data quality, expand emission coverage, and integrate climate considerations into operational decision-making.

Highlights



EMISSION PROFILE

Total GHG
Emissions

REPORT PREPARED BY

Emissions (DG +
Transport)

389.1
tCO₂e

Microcosm Sustainability Intelligence
(Electricity)

1364.7
tCO₂e

A long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance.

1013.76